
BIS638: Final Project

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Acute Respiratory Distress Syndrome in ICU Patients

- Demographic and baseline features associated with developing ARDS and its severity
 - Features of interest: race, gender, age, insurance, tobacco use, assisted living, living status with family, alcohol use
 - Hypothesis:
 - Older patients and tobacco have higher risk in developing ARDS¹
 - Gender association - mixed results²
 - Insurance - Medicaid/Medicare > private

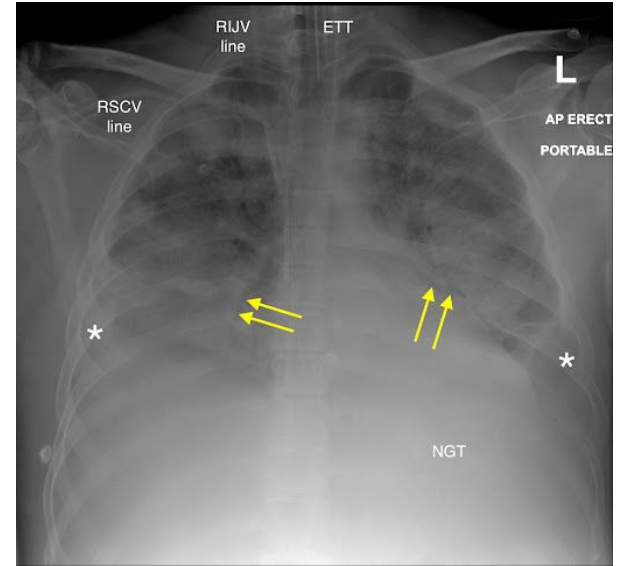
1. Brown R, McKelvey MC, Ryan S, Creane S, Linden D, Kidney JC, McAuley DF, Taggart CC, Weldon S. The Impact of Aging in Acute Respiratory Distress Syndrome: A Clinical and Mechanistic Overview. Front Med (Lausanne). 2020 Oct 26. 2. McNicholas BA, Madotto F, Pham T, Rezoagli E, Masterson CH, Horie S, Bellani G, Brochard L, Laffey JG; LUNG SAFE Investigators and the ESICM Trials Group. Demographics, management and outcome of females and males with acute respiratory distress syndrome in the LUNG SAFE prospective cohort study. Eur Respir J. 2019 Oct 17

Acute Respiratory Distress Syndrome in ICU Patients

- Stratified analysis
 - Association of ARDS severity with in-hospital death
 - In hospital death stratified by different ICU wards (first-care) among ARDS patients
- Trajectory analysis
 - Oxygenation evolve in the first 48 hours after ARDS onset
 - How do different trajectory patterns relate to outcomes (worsening, stable, improving)?

Acute Respiratory Distress Syndrome (ARDS)

- Have clear precipitant e.g., pneumonia, sepsis, trauma/burns, CVT surgery
- Progressive dyspnea (> 1wk)
- Hypoxemic respiratory failure (PaO₂:FiO₂ ratio <300)
- Bilateral alveolar opacities on chest imaging



For simplicity: PaO₂:FiO₂ ratio <300, assume intubation

ARDS - Berlin Criteria and Severity

Table 2: Berlin criteria for diagnosis and prognostication of ARDS.

1. Onset within 7 days after known precipitant (see table 1) or new/progressive respiratory symptoms.
2. Bilateral Opacities on chest X-ray or chest CT.
3. Respiratory distress not due to CHF or volume overload
4. P:F ratio* with PEEP ≥ 5 cmH₂O
 - P:F of 201 - 300; considered mild ARDS with 27% mortality
 - P:F of 101 - 200; considered moderate ARDS with 32% mortality
 - P:F ≤ 100 ; considered severe ARDS with 45% mortality

* Ratio of PaO₂: FiO₂. PaO₂ obtained from ABG or estimated with a pulse oximeter. To determine FiO₂ from litres per minute (LPM) of O₂ use $\text{FiO}_2 = 21\% + (4\% \text{ LPM})$.

Adapted from Thompson, et al 2017, and Fan et al 2018.

Demographic and Baseline Features

- Selected Patients:
 - Sepsis variations and underlying infections (for ICD 9 and 10), such as:
 - Sepsis (995.91)
 - Severe Sepsis (995.92)
 - Extracted PaO_2 (220224) & FiO_2 (223835)
- Patient Features:
 - Insurance, language, marital_status, race, etc.
 - Tobacco use (227687), Alcohol use (225106), living status (225072), etc.

Methods

- Determination of ARDS

- For each ICU stay (grouped by stay_id)

$$ARDS = 1(\frac{(PaO_2)}{(FiO_2)} \leq 300)$$

- Patients with missing PaO_2 and FiO_2 are considered as not suffering from ARDS

- PaO_2 and FiO_2 assumed to be a standard procedure for any patient suspected of having ARDS

- Model

- Random Forest found to be the best model
 - Accuracy: 0.7351
 - ROC-AUC: 0.7889

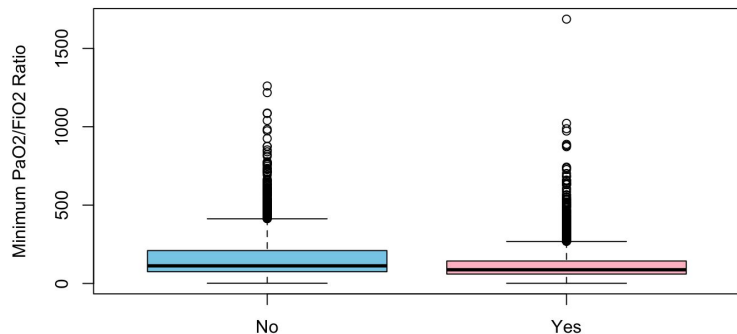
Results

- Top Ranked feature importance
 - Anchor Age - older age is associated with higher risk and mortality of ARDS.¹ Patients with ARDS often carry multiple risk factors and comorbidities, some of which may be influenced by age themselves, which makes isolating the influence of age particularly challenging
 - Gender - Males tend to have more prevalence of ARDS, but the care outcome differences between 2 genders are not significant. It might be likely that male patients are often admitted to ICU than female.²
 - Insurance - Medicare is the U.S. federal health insurance program for people age 65 and older among other factors, showing that adult people/people with disabilities tend to be at risk with ARDS than younger adults/healthy people.

1. Brown R, McKelvey MC, Ryan S, Creane S, Linden D, Kidney JC, McAuley DF, Taggart CC, Weldon S. The Impact of Aging in Acute Respiratory Distress Syndrome: A Clinical and Mechanistic Overview. Front Med (Lausanne). 2020 Oct 26. 2. McNicholas BA, Madotto F, Pham T, Rezoagli E, Masterson CH, Horie S, Bellani G, Brochard L, Laffey JG; LUNG SAFE Investigators and the ESICM Trials Group. Demographics, management and outcome of females and males with acute respiratory distress syndrome in the LUNG SAFE prospective cohort study. Eur Respir J. 2019 Oct 17

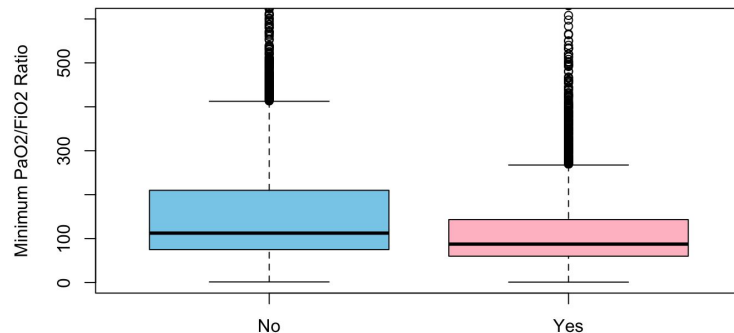
ARDS Severity Distribution and In Hospital Death %

PF Ratio by In-Hospital Death



In-Hospital Death

PF Ratio by In-Hospital Death



In-Hospital Death

ards_severity <chr>	total_admissions <chr>	inhospital_death_pct <chr>
Severe	3612 (52.4%)	49.7%
Moderate	1923 (27.9%)	38.1%
Mild	698 (10.1%)	30.9%
No ARDS	654 (9.5%)	25.1%

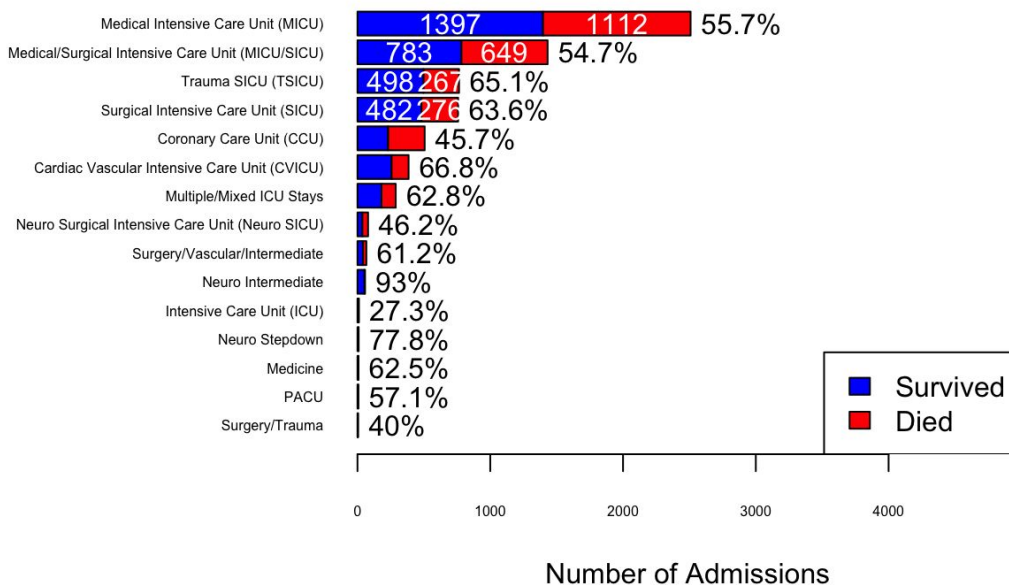
ARDS Severity and Ventilation Care In Hospital Death %

ards_severity <fctr>	ventilation_status <fctr>	inhospital_death_pct <chr>	total_admissions <chr>
Severe	Yes	31.7%	2256 (32.8%)
Severe	No	0.0%	2 (0.0%)
Severe	Mixed	56.4%	165 (2.4%)
Severe	NA	82.9%	1189 (17.3%)
Moderate	Yes	21.5%	1268 (18.4%)
Moderate	No	NA%	0 (0.0%)
Moderate	Mixed	50.0%	52 (0.8%)
Moderate	NA	71.8%	603 (8.8%)

ards_severity <fctr>	ventilation_status <fctr>	inhospital_death_pct <chr>	total_admissions <chr>
Mild	Yes	14.1%	461 (6.7%)
Mild	No	NA%	0 (0.0%)
Mild	Mixed	26.7%	15 (0.2%)
Mild	NA	66.2%	222 (3.2%)
No ARDS	Yes	12.6%	429 (6.2%)
No ARDS	No	NA%	0 (0.0%)
No ARDS	Mixed	25.0%	8 (0.1%)
No ARDS	NA	49.8%	217 (3.2%)

First Care Unit Type Survival Rate for Sepsis

Admissions by First Care Unit and In-Hospital Death



Trajectory Analysis: Objective

Building trajectories

- Used $\text{PaO}_2/\text{FiO}_2$ values from ARDS onset up to 48 hours while patients were on mechanical ventilation
- Placed each value on a regular 4 hour timeline from 0 to 48 hours
- Kept ICU stays with at least four $\text{PaO}_2/\text{FiO}_2$ measurements within this 48 hour window

Trajectory features and groups

- Defined change as $\text{PaO}_2/\text{FiO}_2$ at the last time point minus the first time point
- Assigned stays with change less than or equal to zero to a True worsening group
- Applied k means clustering with two clusters to the remaining trajectories and labeled them Improving slow and Improving fast

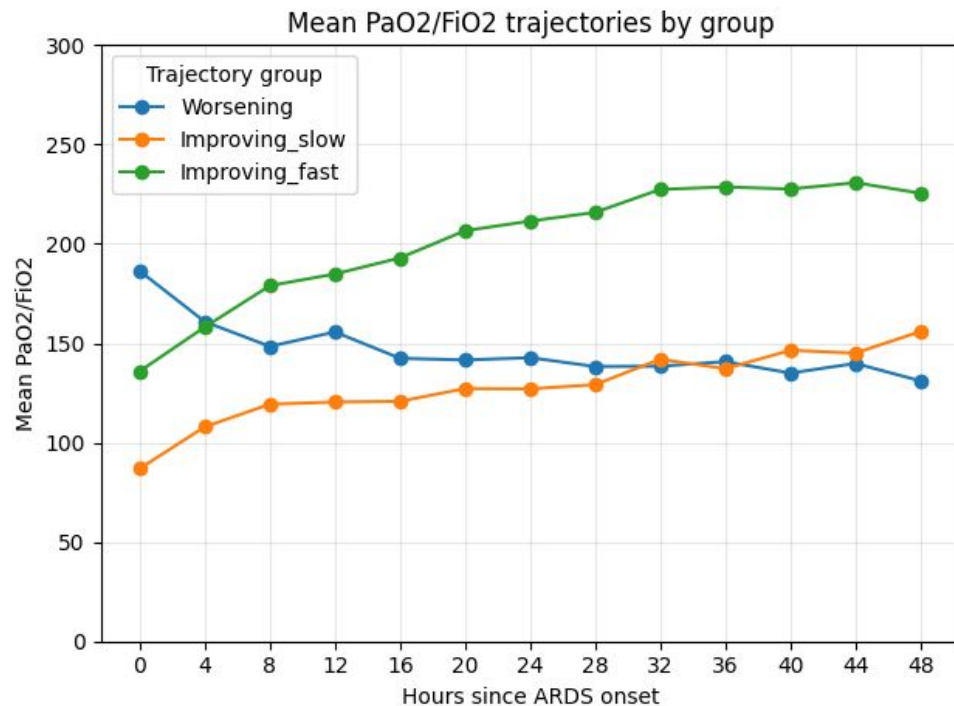
Link to outcomes

- Summarized mortality, median ventilation days, and minimum $\text{PaO}_2/\text{FiO}_2$ by trajectory group

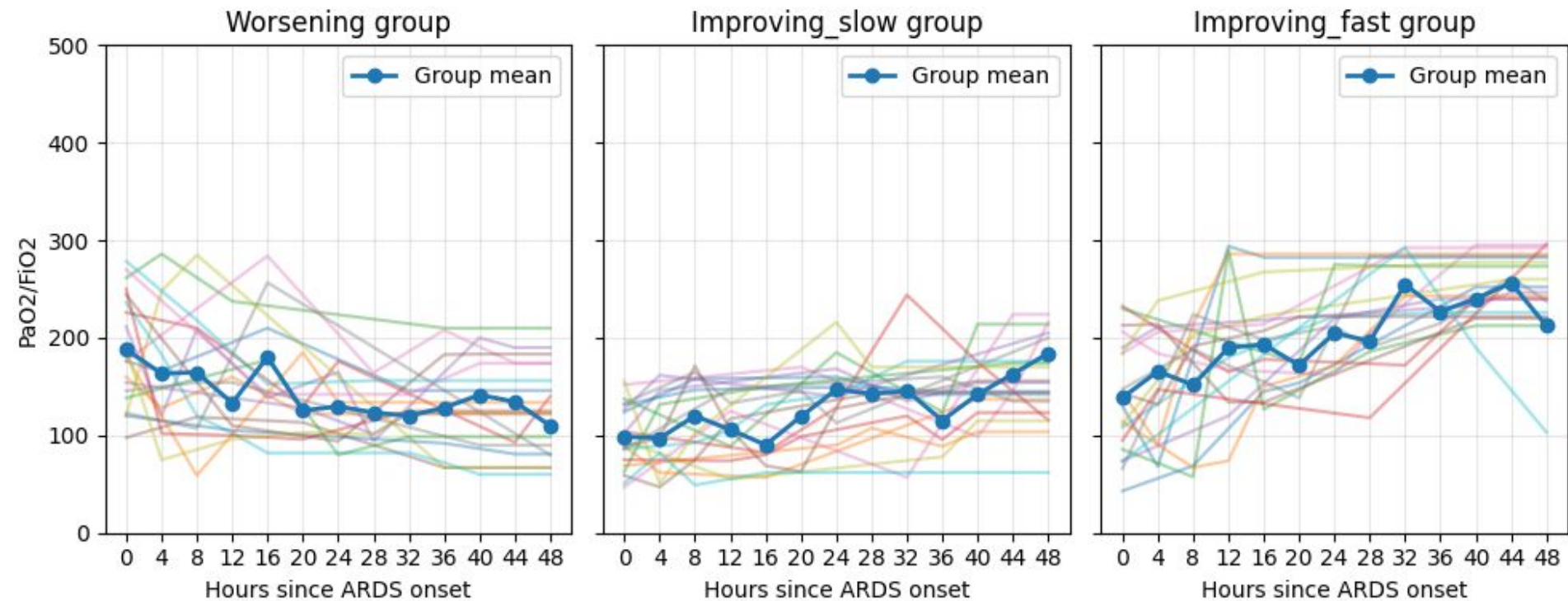
Trajectory Results Across Groups

PaO₂/FiO₂ trajectories after ARDS onset

- Three distinct patterns of oxygenation
- Worsening group starts with relatively higher PaO₂/FiO₂ then drifts down and remains low
- Improving slow group starts low and shows only gradual improvement over time
- Improving fast group shows a rapid rise in PaO₂/FiO₂ and reaches the best oxygenation levels by 48 hours

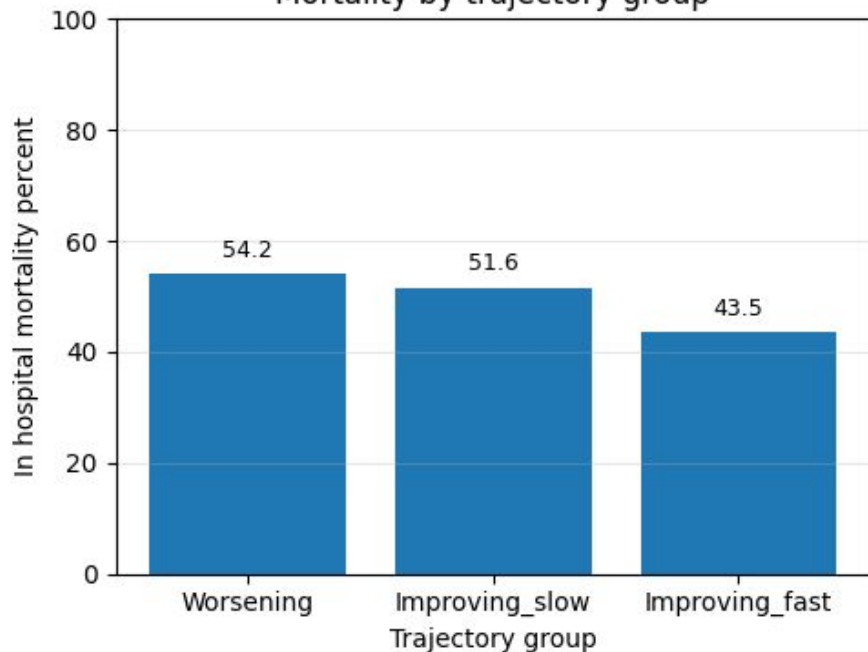


Trajectory Results Within Group

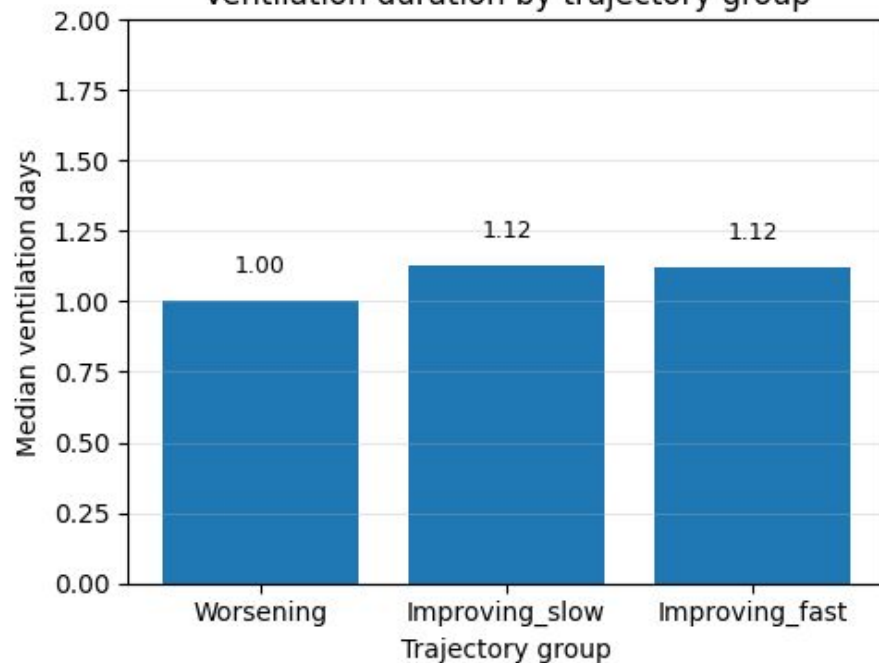


Trajectory and Outcomes

Mortality by trajectory group



Ventilation duration by trajectory group



Q&A

Contributions

- Andrew - stratified analysis
- Mengyao - trajectory analysis
- Nat - formulate project questions, variable selection, data quality check, interpretation
- Vincent - demographic and baseline features analysis